MYP eAssessment

Students seeking **IB MYP course results** for MYP sciences complete an on-screen examination in which they can demonstrate their achievement of subject-group objectives. Successful results can contribute to students' attainment of the **IB MYP certificate**.

Optional eAssessment in sciences is offered in biology, chemistry, physics and integrated sciences and is assessed by on-screen examination. This verification of learning assures accurate and consistently applied standards.

Year 5 MYP eAssessment candidates may be registered for one, or more, discrete science subjects, or may be registered for integrated sciences. A candidate cannot register for an additional discrete science subject, such as biology, physics, or chemistry, when registered for integrated sciences.

Topic lists for sciences

For the purpose of external assessment, the MYP identifies a range of subject-specific topics that constitute one of the variables that authors consider when they create on-screen examinations. These topics are at a lower level of specification than the formal syllabus of a similar subject in the IB Diploma Programme, and they leave considerable leeway for schools to develop their own written curriculum according to MYP requirements.

These topics define the examinable subject matter for MYP on-screen examinations. In their local development of the MYP curriculum, schools are not limited to these topics. This list does not constitute an exclusive IB-approved curriculum for MYP years 4–5.

Biology

- Cells (tissues, organs, systems, structure and function; factors affecting human health; physiology; vaccination)
- Organisms (habitat, ecosystems, interdependency, unity and diversity in life forms; energy transfer and cycles [including nutrient, carbon, nitrogen]; classification)
- Processes (photosynthesis, cell respiration, aerobic and anaerobic, word and chemical equations)
- Metabolism (nutrition, digestion, biochemistry and enzymes; movement and transport, diffusion; osmosis; gas exchange; circulation, transpiration and translocation; homeostasis)
- Evolution (life cycles, natural selection; cell division, mitosis, meiosis; reproduction; biodiversity; inheritance and variation, DNA and genetics)
- Interactions with environment (tropism, senses, nervous system, receptors and hormones)
- Interactions between organisms (pathogens/parasites, predator/prey, food chains and webs; competition, speciation and extinction)
- Human interactions with environments (human influences, habitat change or destruction, pollution/conservation; overexploitation, mitigation of adverse effects)

- Biotechnology (genetic modification, cloning; ethical implications, genome mapping and application, 3D tissue and organ printing)

Chemistry

- Periodic table (metals and non-metals; transition metals, noble gases, trends, periods, groups)
- International Union of Pure and Applied Chemistry (IUPAC naming and classification of: alkanes, alkenes, alcohols, carboxylic acids and esters; structural formulas)
- The atmosphere (characteristics of gases; atmospheric composition, testing and treatment; extraction, emission and environmental implications)
- Matter (states and properties of matter; particle/kinetic theory, diffusion; atomic structure [including isotopes]; electron configuration and valency)
- Pure and impure substances (types of mixtures [solutions, oils, alloys, emulsions]; separation techniques, including: filtration, distillation [including crude oil], chromatography)
- Bonding (structure and bonding, properties, chemical formulas, chemical reactions and the conservation of mass; balancing equations, the mole concept and chemical calculations; reaction kinetics [rates, and factors affecting rates/collision theory]; equilibria/reversible reactions; energy changes in reactions, endo- and exothermicity; combustion of fuels)
- Types of chemical reaction (acids and bases, neutral solutions, acid/base reactions, pH and indicators, formation of salts, uses of salts; redox reactions, reactivity series; extraction of metals, and corrosion, electrochemical cells)

Physics

- Forces and energy (measurement in science; states and properties of matter, kinetic theory, density; forces and effects of forces; forces and motion, speed, motion graphs, Newton's laws; pressure; work and power, efficiency; gravity and gravitational fields; energy sources and resources, fuels and environmental impact; energy transfer and transformation, conservation of energy)
- Electromagnetism (magnetism, electric and magnetic fields; static electricity; electromagnetic forces and induction, AC and DC; current, voltage, power, generation and transmission of electricity; electric circuits)
- Astrophysics (the solar system, planets and satellites, the Big Bang theory)
- Heat, light and sound (thermal physics; heat transfer, condensation and evaporation)
- Waves (longitudinal and transverse waves, sound waves; wave phenomena including reflection, refraction, diffraction; wave equation; electromagnetic spectrum, imaging and applications)
- Atomic physics (atomic structure, particles, charges and masses; radioactivity, decay and half-life, forms of radiation; uses and dangers)

Integrated sciences

- Atoms (atomic structure and electron configuration)
- Bonding (word and chemical reactions and formulas; acids, bases and pH)
- Cells (tissues, organs and systems; cell division; reproduction)
- Electromagnetism (magnetism, magnetic fields; electric circuits)
- Forces and energy (motion, motion graphs, Newton's laws; energy transfer and transformation)
- Fuels (combustion)

- Interactions between organisms (food chains and webs)
- Matter (particles and kinetic theory)
- Metabolism (digestion, gas exchange)
- Periodic table (trends, periods, groups)
- Systems (photosynthesis and respiration)
- Waves (longitudinal and transverse waves, sound waves; wave phenomena including reflection, refraction, diffraction; wave equation; electromagnetic spectrum)

Sciences examination blueprint

MYP on-screen examinations are constructed as a series of tasks that sample, simulate or replicate internal assessment practices. The assessments follow an agreed structure that provides a clear framework for developing each examination. The distribution of marks within each eAssessment may vary by no more than three marks from those displayed in the blueprint.

As part of an ethical assessment model, these assessment blueprints ensure consistency and transparency, and they guarantee a balanced approach in measuring students' achievement with respect to MYP objectives. MYP on-screen examination blueprints document the close connection of large-scale assessment with subject-group objectives, classroom learning engagements and the programme's rigorous internal assessment requirements.

These blueprints enable teachers and students to review the nature and purpose of MYP eAssessment. They provide an important resource for helping students to prepare for on-screen examinations, focusing attention on subject-group criteria and assessment strategies in each subject group.

Overview

The following table illustrates how on-screen examinations in the sciences assessment are structured.

Task	Marks	Main criteria assessed	Criterion marks
Knowing and understanding	25	A	25
Investigation skills	50	B	25
		C	25
Applying science	25	D	25
	100		